Lecture 6 Question 6

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We need a wrapper for bio3d that (1) retrieves one or more PDB files, (2) filters each to a specified chain and atom type, (3) generates a B-factor profile plot for the selected structure, and (4) returns the plotted output.

First install the package

It provides functions to read PDB files (read.pdb()), subset them (trim.pdb()), and visualize B-factor profiles (plotb3()).

```
install.packages("bio3d", repos = "<https://cloud.r-project.org>") # install.packages("bio3d")
```

```
## Installing package into 'C:/Users/chenr/AppData/Local/R/win-library/4.5'
## (as 'lib' is unspecified)

## Warning: unable to access index for repository <https://cloud.r-project.org>/src/contrib:
##   cannot open URL '<https://cloud.r-project.org>/src/contrib/PACKAGES'

## Warning: package 'bio3d' is not available for this version of R
##
## A version of this package for your version of R might be available elsewhere,
## see the ideas at
## https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages

## Warning: unable to access index for repository <https://cloud.r-project.org>/bin/windows/contrib/4.5
##   cannot open URL '<https://cloud.r-project.org>/bin/windows/contrib/4.5/PACKAGES'
```

```
library(bio3d)
```

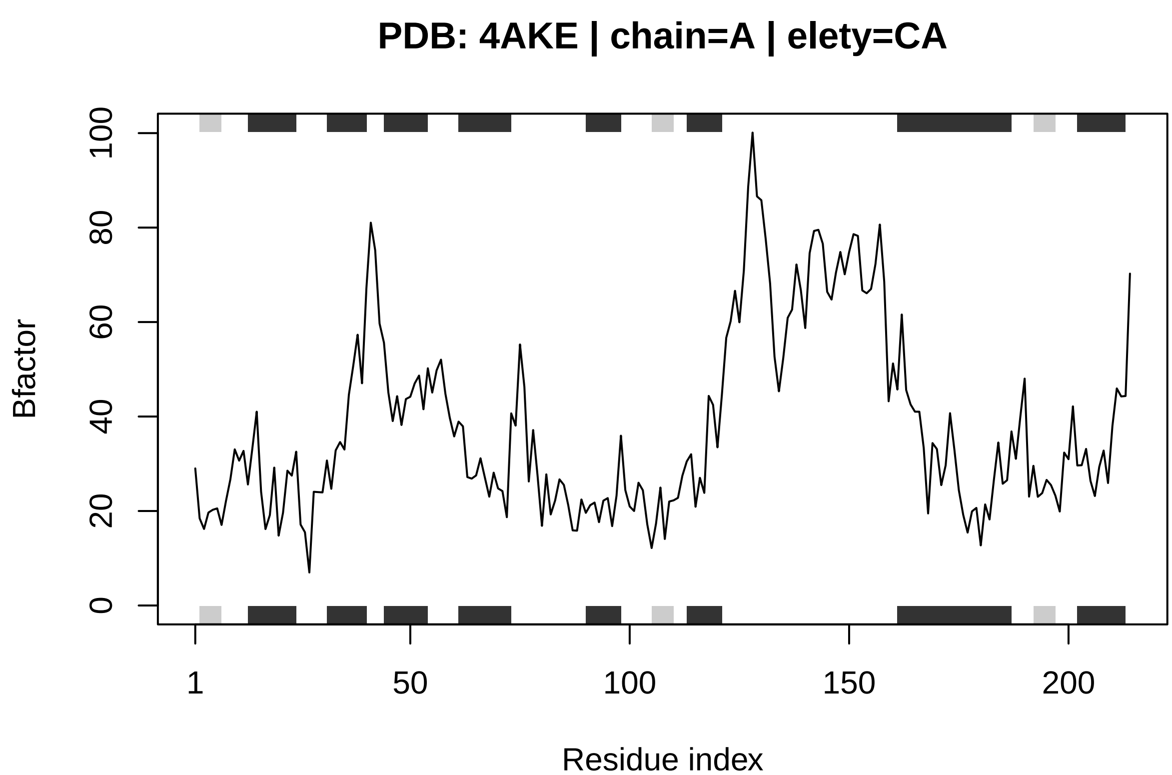
Generalizing B-factor analysis to work with any collection of PDB structures, allowing B-factors to be plotted for a single selected protein from a given set of PDB IDs.

```
plot_protein <- function(pdb_ids, which = 1, chain = "A", elety = "CA") {
  stopifnot(length(pdb_ids) > 0)  # Require at least one PDB ID
  stopifnot(which >= 1 && which <= length(pdb_ids))  # Validate index

  pdbs <- lapply(pdb_ids, read.pdb)  # Load PDB files from RCSB

  trims <- lapply(pdbs, function(p) {
    trim.pdb(p, chain = chain, elety = elety)
    # Retain only the specified chain and atom type (e.g., CA)
  })

  sel <- trims[[which]]        # Select the desired structure
  b <- sel$atom$b              # Extract B-factor values

  plotb3(
    b,                         # Plot B-factors vs residue index
```

```r
    sse = sel,                  # Add secondary structure annotation
    typ = "l",                  # Line plot
    ylab = "Bfactor",           # Y-axis label
    xlab = "Residue index"      # X-axis label
  )

  title(main = paste0(
    "PDB: ", pdb_ids[which],
    " | chain=", chain,
    " | elety=", elety
  ))

  invisible(list(
    selected_pdb = pdb_ids[which],  # Selected PDB ID
    b = b,                          # B-factor vector
    trimmed = sel                   # Trimmed PDB object
  ))
}
```

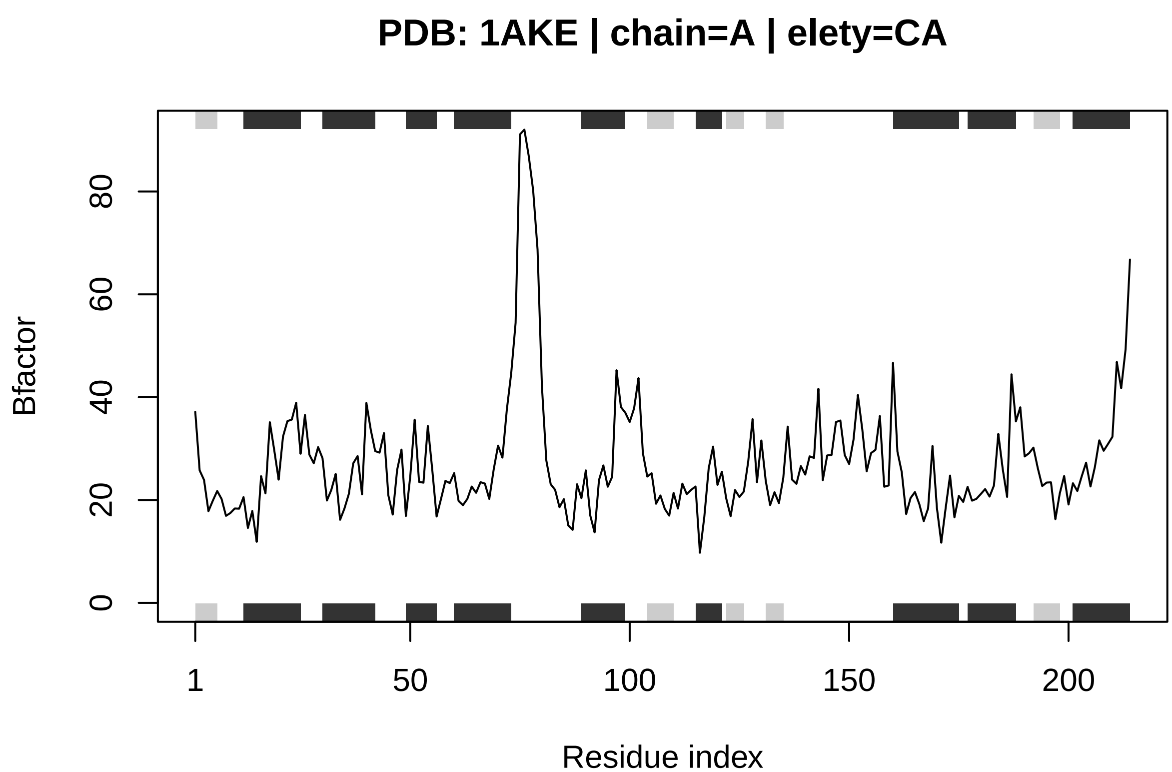
```r
ids <- c("4AKE","1AKE","1E4Y") # Define the set of PDB IDs to analyze
plot_protein(ids, which = 1) # Plot the first protein's B-factors
```

```
##   Note: Accessing on-line PDB file
##   Note: Accessing on-line PDB file
##    PDB has ALT records, taking A only, rm.alt=TRUE
##   Note: Accessing on-line PDB file
```



```r
plot_protein(ids, which = 2)
```

```
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/4AKE.pdb exists. Skipping download

##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/1AKE.pdb exists. Skipping download

##    PDB has ALT records, taking A only, rm.alt=TRUE
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/1E4Y.pdb exists. Skipping download
```



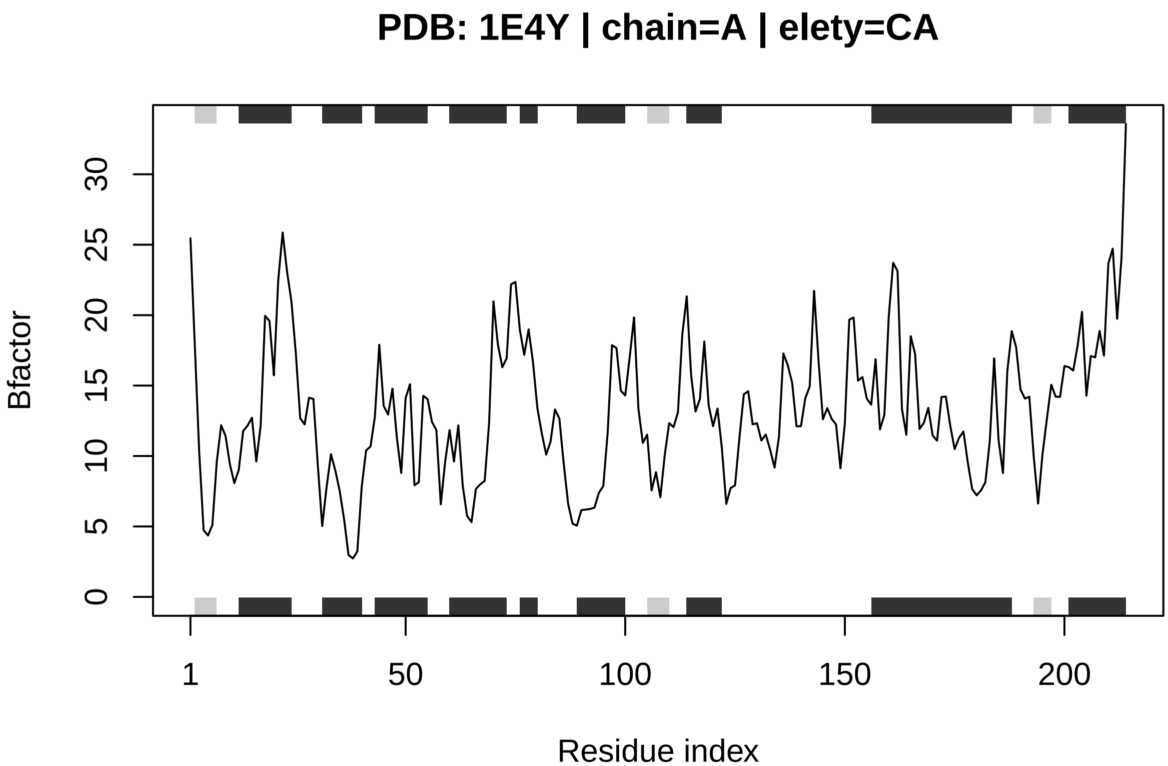

```r
# Plot the second protein's B-factors
```

```r
plot_protein(ids, which = 3) # Plot the third protein's B-factors
```

```
##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/4AKE.pdb exists. Skipping download

##   Note: Accessing on-line PDB file

## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/1AKE.pdb exists. Skipping download
```

```
##    PDB has ALT records, taking A only, rm.alt=TRUE
##   Note: Accessing on-line PDB file
```

```
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
## C:\Users\chenr\AppData\Local\Temp\RtmpqIF3E7/1E4Y.pdb exists. Skipping download
```



R uses chain = "A" and elety = "CA" as default arguments, so you only need to specify them if you want different values.